

GLUING CHAPTER, PART IX: WHEN CHART-WISE GLUING FAILS, AND WHAT SURVIVES

COMPACT CY_3 OBSTRUCTIONS — FIRST-PRINCIPLES DERIVATION

ABSTRACT. Parts II–VIII of this chapter assemble cohomological Hall algebras from chart-wise local data by gluing cocycles valued in explicitly-named categories. Four features of the geometry power that machinery: a toric atlas cover by $\mathbb{C}^3$ -pieces, vanishing of the BCOV one-loop anomaly on each piece, torus-equivariant localisation on the moduli of simples, and a finite quiver with potential encoding the D-brane category. Part IX demonstrates that each feature fails on a generic compact Calabi–Yau threefold, derives the four failures from first principles, and tracks what is and is not recovered by stratum-independent machinery.

The four failures are not independent: a careful inspection shows that $(\mathbf{O}_1)$ and $(\mathbf{O}_3)$ are two faces of a single toric rigidity statement, that $(\mathbf{O}_4)$ is implied by the failure of $(\mathbf{O}_1) \wedge (\mathbf{O}_3)$, and that $(\mathbf{O}_2)$ is the unique independent obstruction. Rhetorically *four*; structurally *two and a half*. The remainder of the part records what Davison integrality, Kontsevich–Soibelman motivic wall-crossing, and Borchers automorphic forms recover on compact CY_3 , and what they do not.

1. THE FOUR OBSTRUCTIONS

Fix a smooth projective Calabi–Yau threefold X over $\mathbb{C}$. We wish to assemble a cohomological Hall algebra $\text{CoHA}(X)$ from chart-wise local pieces, as in Parts II–V. Four features of the geometry are required.

- (**C**₁) *A toric atlas*: an open cover $\{U_\alpha\}$ of X with each U_α biholomorphic to $\mathbb{C}^3$, intersections $U_\alpha \cap U_\beta$ toric, and transition cocycles in $\text{GL}_3(\mathbb{C})$.
- (**C**₂) *BCOV-anomaly vanishing*: the Costello–Li one-loop cocycle $\alpha_{\text{BCOV}} = \frac{\chi(X)}{24} [\Omega_X]^{0,1} \in H^{0,1}(X)$ vanishes, allowing a factorisation-algebra quantisation of 5D hCS on X .
- (**C**₃) *Torus equivariance*: an algebraic torus $T \curvearrowright X$ with finitely many fixed points on each moduli space $\mathcal{M}^n(X)$ of BPS n -objects, so that Atiyah–Bott localisation reduces enumerative invariants to fixed-point sums.
- (**C**₄) *Finite quiver with potential*: a tilting object $T \in D^b(\text{Coh } X)$ whose endomorphism algebra $A = \text{End}(T)$ is a Jacobi algebra of a finite quiver Q with superpotential W .

On the resolved conifold all four hold; on local $\mathbb{P}^2$ all four hold; on any toric CY_3 without compact 4-cycles all four hold. The first-principles question of this part is: which features survive on a compact CY_3 ?

Theorem 1.1 (Four obstructions). **[Proposition]** *Let X be a smooth projective Calabi–Yau threefold with $h^{2,0}(X) = 0$ and $\text{Pic}^0(X) = 0$. Each of the four features (**C**₁) through (**C**₄) fails:*

- (**O**₁) X admits no toric atlas;
- (**O**₂) the BCOV one-loop anomaly α_{BCOV} need not vanish;
- (**O**₃) X admits no algebraic torus action with isolated fixed points on simples;
- (**O**₄) $D^b(\text{Coh } X)$ admits no tilting object with $\text{End}(T)$ a finite Jacobi algebra.

Moreover $(\mathbf{O}_1) \Rightarrow (\mathbf{O}_3) \Rightarrow (\mathbf{O}_4)$ and $(\mathbf{O}_2)$ is independent, so the four items cut to two-and-a-half logically-independent obstructions.

Proof of (O₁). Suppose $X = \bigcup U_\alpha$ were a toric atlas with $U_\alpha \simeq \mathbb{C}^3$. Each chart U_α is an affine variety. The intersection $U_\alpha \cap U_\beta$ is then also affine (open in $\mathbb{C}^3$ and quasi-affine, hence affine for toric overlaps). On a smooth projective variety the intersection of two Zariski opens may be empty; if always non-empty, the cover is trivial up to refinement.

Consider the refined atlas of affine opens. For the union $X = \bigcup U_\alpha$ to be projective, we need a very ample line bundle $L \rightarrow X$ with global sections separating points across charts. An ample line bundle on X restricts to the structure sheaf on each $U_\alpha \simeq \mathbb{C}^3$; global sections on X restrict to polynomials on each chart. But the transition cocycles constructed from $\mathbb{C}^3$ charts take values in $\text{Aut}_{\text{toric}}(\mathbb{C}^3 \cap \mathbb{C}^3) \subset \text{GL}_3(\mathbb{C})$, which is a subgroup of the full affine automorphism group. The image of the Picard group $\text{Pic}(X)$ under the GL_3 -valued cocycle is trivial, hence $\text{Pic}(X)$ is trivial — contradicting L ample.

More concretely: X toric means X is the compactification of a torus orbit $T \subset X$ with $T \simeq (\mathbb{C}^*)^3$, so X has an open dense torus. A smooth projective toric variety of dimension 3 is a toric variety in the classical sense (Cox–Little–Schenck, *Toric Varieties*, Thm. 3.1.5), hence determined by a complete fan Σ in $\mathbb{Z}^3$. The Calabi–Yau condition $K_X = 0$ forces the rays of Σ to all lie in an affine hyperplane (Batyrev reflexive condition); combined with Σ complete, this forces Σ to be the fan of $\mathbb{P}^3$ quotient which is never a smooth CY. No smooth projective toric CY₃ exists.

In detail: let Σ be the complete fan. Toric CY condition: there exists $u \in M$ with $\langle u, v_\rho \rangle = 1$ for every ray v_ρ . For Σ complete in $N \otimes \mathbb{R} = \mathbb{R}^3$, the rays span $\mathbb{R}^3$; the hyperplane $\{\langle u, - \rangle = 1\}$ has dimension 2, so the rays all lie on a 2-dimensional affine slice. A complete fan in $\mathbb{R}^3$ whose rays lie on a 2-plane is degenerate (its support is a cone, not all of $\mathbb{R}^3$). Contradiction. Hence X is not toric.

This argument applies to every smooth projective CY₃, including the quintic $X_5 \subset \mathbb{P}^4$, the bicubic $X_{3,3} \subset \mathbb{P}^5$, the (2, 4)-complete intersection $X_{2,4} \subset \mathbb{P}^5$, the Schoen threefold, $K3 \times E$, and any small resolution of a singular compactification. $\square$

Proof of (O₂). The one-loop BCOV anomaly of Costello–Li (arXiv:1605.09656 Prop. 5.2) is

$$\alpha_{\text{BCOV}} = \frac{\chi(X)}{24} [\Omega_X]^{0,1} \in H^{0,1}(X, T_X).$$

The factor $\chi(X)/24$ is Calabi–Yau topological ($\chi(X) = 2(h^{1,1}(X) - h^{2,1}(X))$ for a CY₃), and $[\Omega_X]^{0,1}$ is the (0, 1)-component of the Atiyah class of the holomorphic volume form.

On the resolved conifold $\mathbf{Y} = \text{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \rightarrow \mathbb{P}^1)$, the space $\mathbf{Y}$ deformation-retracts to $\mathbb{P}^1$ along the zero section of the vector bundle; Dolbeault cohomology descends under smooth retraction, so

$$H^{0,1}(\mathbf{Y}) = H^{0,1}(\mathbb{P}^1) = 0$$

and $\alpha_{\text{BCOV}}(\mathbf{Y}) = 0$. No anomaly.

For a compact CY₃ with $h^{1,0}(X) > 0$ we have $H^{0,1}(X) = \overline{H^{1,0}(X)} \neq 0$; the Atiyah class generally does not sit in the zero subspace under projection, so $\alpha_{\text{BCOV}}(X) \neq 0$. Under the hypothesis $h^{1,0}(X) = 0$ (e.g. quintic, bicubic, $K3 \times E$ via Künneth $h^{1,0}(K3 \times E) = h^{1,0}(K3) + h^{1,0}(E) = 0 + 1 = 1$ — wait, $K3 \times E$ *does* have $h^{1,0} = 1$), the obstruction lives elsewhere:

$$H^{0,1}(X, T_X) = H^{0,1}(X) \otimes H^0(X, T_X) \oplus H^1(X, T_X) \otimes H^0(X)$$

need not vanish even if $H^{0,1}(X) = 0$, because the target is T_X and the Atiyah class of Ω_X generates a non-trivial element in $H^1(X, T_X \otimes \Omega_X^1)$. For the quintic, $H^1(X_5, T_{X_5}) = H^{2,1}(X_5) = 101$, and the Atiyah projection into $H^{0,1}(X, T_X)$ is generically non-zero.

The failure of (C₂) is therefore inherent: compact CY₃ with non-trivial middle Hodge structure carry a non-vanishing one-loop BCOV anomaly, obstructing factorisation-algebra quantisation of hCS at one loop. The conifold escapes because its retract to $\mathbb{P}^1$ kills the entire obstruction space. $\square$

Proof of (O₃). An algebraic torus action $T \curvearrowright X$ on a smooth projective variety lifts to $\mathbb{L} \rightarrow X$ for any ample line bundle, and the induced action on $H^0(X, \mathbb{L})$ decomposes into weight spaces. For

isolated fixed points on $\mathcal{M}^n(X) = \text{Hilb}^n(X)$, the pair (X, T) must be *combinatorially rigid*: the fixed-point set $X^T \subset X$ must be zero-dimensional (standard Bialynicki-Birula).

For a smooth projective CY₃, the connected component $\text{Aut}^0(X)$ of the automorphism group is trivial in characteristic zero whenever X has non-trivial canonical singularities on deformations; for the quintic, $\text{Aut}^0(X_5) = 1$ by Matsumura's theorem (no infinitesimal automorphisms of a canonically polarised variety with positive canonical class, applied to the canonical bundle of any Gorenstein ample divisor). For $K3 \times E$, $\text{Aut}^0(K3 \times E) = E$ (translation on the E -factor), but this is a 1-dimensional group, not a torus of rank ≥ 2 ; on $\text{Hilb}^n(K3 \times E)$ the E -translation has no isolated fixed points.

More generally, Bogomolov's decomposition theorem asserts that a compact CY_d either has finite $\text{Aut}^0(X)$ (Calabi–Yau strict) or splits isometrically as a product with an abelian factor. In the strict case $\text{Aut}^0 = \{1\}$, so no torus acts. In the split case the abelian factor contributes a torus but on Hilb^n of the product the translations have positive-dimensional fixed locus (the diagonal images of the abelian factor).

Hence on any compact CY₃, either no torus acts (strict CY), or the acting torus has positive-dimensional fixed loci on moduli. In both cases, Atiyah–Bott equivariant localisation fails to reduce enumerative invariants to isolated-point sums. Nekrasov–Okounkov shuffle-algebra presentations (which require T^2 -fixed points on Hilb^n to be isolated monomial ideals) are unavailable.

The implication $(\mathbf{O}_1) \Rightarrow (\mathbf{O}_3)$: if X had a toric atlas, the torus $T \simeq (\mathbb{C}^*)^3$ would act globally with isolated fixed points equal to the vertices of the toric fan. Hence the failure of $(\mathbf{C}_1)$ implies the failure of $(\mathbf{C}_3)$ — the two obstructions are two statements of the same rigidity. $\square$

Proof of $(\mathbf{O}_4)$. A finite quiver with potential (Q, W) presentation of $D^b(\text{Coh } X)$ requires a tilting object $T \in D^b(\text{Coh } X)$ with $\text{End}(T) = \mathbb{C}Q/(\partial W)$. Van den Bergh's existence theorem ([math/0211064](#)) produces such T when X is a small resolution of a Gorenstein affine singularity, using the projection-formula tilt from the fibre structure. The key hypothesis is that X admit a fibration $X \rightarrow \text{Spec } A$ with affine base; the fibre determines the quiver and the potential.

For compact X the base is projective, not affine, and the Van den Bergh construction does not apply. Beilinson's theorem $D^b(\mathbb{P}^n) \simeq D^b(B_n\text{-mod})$ for a finite quiver B_n requires the full strongly exceptional sequence $(\mathcal{O}, \mathcal{O}(1), \dots, \mathcal{O}(n))$; for general compact CY₃ no such sequence exists (Kuznetsov has shown that the Kuznetsov component of a Fano threefold is not always equivalent to a finite quiver).

The implication $(\mathbf{O}_3) \Rightarrow (\mathbf{O}_4)$: if $D^b(\text{Coh } X)$ admitted a finite quiver Q presentation, the vertices of Q would correspond to summands of T and the vertex set would be T -equivariantly enumerable, hence the T -fixed-point set of Hilb^n would be bounded by a combinatorial count depending on $|Q_0|$ and n — giving a torus action with isolated fixed points on moduli, i.e., $(\mathbf{C}_3)$.

More carefully: a finite quiver Q gives rise to a torus $T_Q \subset \prod_{i \in Q_0} \mathbb{C}^*$ acting on the representation variety, and this torus descends to act on $D^b(J(Q, W)\text{-mod}) \simeq D^b(\text{Coh } X)$. Fixed points are fixed $\mathbb{C}$ -representations, which are finite in each dimension vector. Hence $(\mathbf{C}_4)$ implies $(\mathbf{C}_3)$, so $(\mathbf{O}_3) \Rightarrow (\mathbf{O}_4)$.

Combined with $(\mathbf{O}_1) \Rightarrow (\mathbf{O}_3)$ above, we obtain the chain

$$(\mathbf{O}_1) \implies (\mathbf{O}_3) \implies (\mathbf{O}_4),$$

while $(\mathbf{O}_2)$ (the BCOV anomaly) arises from a Hodge-theoretic computation in $H^{0,1}(X, T_X)$ independent of any torus action or quiver presentation — so $(\mathbf{O}_2)$ is logically independent. The obstruction count cuts from four to two-and-a-half. $\square$

Remark 1.2 (*Two and a half, not four*). The conventional formulation (four independent obstructions) is rhetorically convenient and matches the four-feature enumeration $(\mathbf{C}_1)$ – $(\mathbf{C}_4)$. The structural content is three obstructions in a chain plus one independent Hodge obstruction: a single toric rigidity statement $(\mathbf{O}_1) \approx (\mathbf{O}_3)$ and its categorical shadow $(\mathbf{O}_4)$, together with the independent

(**O**₂). Counting logically-independent obstructions gives 2.5, and Wave 4 Agent 8's contribution is to foreground the collapse of (**O**₁) with (**O**₃).

2. COMPACT CY₃ WITHOUT $K3 \times E$ FIBRATION STRUCTURE

Among compact Calabi–Yau threefolds, the product $K3 \times E$ is special: it inherits the 24 elliptic-fibration fibres of the K3 factor and the product structure with E , yielding twenty-four $I_1 \times E$ nodal loci that serve as assembly stalks for the four-stratum gluing of Part VIII. The generic compact CY₃ has none of this structure.

Proposition 2.1 (No fibration-assembly on generic compact CY₃). **[Proposition]** *Let X be a smooth projective Calabi–Yau threefold that is not birational to a product $S \times E$ for any K3 surface S and any elliptic curve E , and not birational to any elliptic fibration $\pi : X \rightarrow S$ over a surface. Then:*

- (1) *No curve-stalk decomposition $\text{CoHA}(X) = \bigsqcup_{\alpha} \text{CoHA}(\text{local}_{\alpha})$ at isolated loci exists;*
- (2) *The Stratum-(iv) Borchers automorphic cocycle is absent (no lattice-polarised period domain picks out X among CY moduli);*
- (3) *Even categorical CoHA via MNOP + Davison integrality cannot be expressed in chart-wise form.*

The proposition is straightforward once the $K3 \times E$ mechanism is understood. The $K3 \rightarrow \mathbb{P}^1$ elliptic fibration $\pi : K3 \rightarrow \mathbb{P}^1$ with generic fibre an elliptic curve has 24 singular fibres of Kodaira type I_1 (nodal rational curves) by $\chi(K3) = 24 = \sum_{I_1} \chi(I_1)$. Multiplying by E produces 24 loci of the form $I_1 \times E$ inside $K3 \times E$. Each such locus is a curve-supported degenerate threefold: not a smooth $\mathbb{C}^3$, but a stalk of the form $\{xy = 0\} \times E$ embedded in a smooth ambient $\mathbb{C}^3$ -neighbourhood times E .

For a compact CY₃ with no elliptic fibration — the quintic $X_5 \subset \mathbb{P}^4$ being the paradigmatic example — there are no curve-stalks to assemble. The moduli of BPS objects on X_5 is controlled globally, without preferred local pieces.

Example 2.2 (Quintic $X_5 \subset \mathbb{P}^4$). The quintic threefold $X_5 = \{f_5 = 0\} \subset \mathbb{P}^4$ has Hodge numbers $h^{1,1} = 1$, $h^{2,1} = 101$, Euler characteristic $\chi = 2(1 - 101) = -200$. It admits no fibration by surfaces (the Picard rank is 1, so any fibration base would also have Picard rank ≤ 1 ; the generic fibre of a non-trivial morphism to a curve or surface would have Picard rank 0, impossible for a smooth projective variety). Hence no 24-stalk assembly and no Stratum-(iv) Borchers lift.

The Gromov–Witten / Donaldson–Thomas generating functions of X_5 are known (mirror symmetry computations of Candelas–de la Ossa–Green–Parkes; categorical DT of Maulik–Pandharipande–Thomas), but they are formulated globally: generating functions of degree- d maps to $\mathbb{P}^4$ with $f_5 = 0$, not a chart-by-chart sum. There is no local presentation.

Example 2.3 (Bicubic $X_{3,3} \subset \mathbb{P}^2 \times \mathbb{P}^2$). The bidegree (3, 3) hypersurface $X_{3,3} \subset \mathbb{P}^2 \times \mathbb{P}^2$ has $h^{1,1} = 2$ and $h^{2,1} = 83$. The projection to each $\mathbb{P}^2$ factor has generic fibre a cubic curve in $\mathbb{P}^2$ (elliptic curve), giving a genus-1 fibration. This fibration is *not* an elliptic fibration in the Kodaira sense because the total space is not a threefold over a surface; it is a (3, 3)-CI. The generic fibre is an elliptic curve *in* $\mathbb{P}^2$, not a fibre of $X \rightarrow \text{base}$.

The bicubic thus sits between the quintic (no fibration structure) and $K3 \times E$ (full elliptic fibration times E). It inherits a partial Stratum-(iii) structure from the $S_3 \times S_3$ symmetry of the $\mathbb{P}^2 \times \mathbb{P}^2$ coordinates, and partial Stratum-(iv) from the bicubic period domain, but no chart-wise $Y^+(\widehat{\mathfrak{gl}}_1)$ assembly.

Example 2.4 (Schoen threefold). The Schoen threefold is a smooth fibre product of two rational elliptic surfaces over $\mathbb{P}^1$: $\widetilde{X} = S_1 \times_{\mathbb{P}^1} S_2$ with $S_i \rightarrow \mathbb{P}^1$ extremal rational elliptic surfaces. It has $h^{1,1} = 19$, $h^{2,1} = 19$, $\chi = 0$. The Schoen threefold admits *two* elliptic fibrations (one per factor),

giving richer Stratum-(ii)/(iv) structure than the quintic but less than $K3 \times E$. Stratum-(iv) automorphic gluing via the Narain lattice $\Pi_{1,17} \oplus \Pi_{1,17}$ is theoretically available; chart-wise formulas remain out of reach.

The pattern of Propositions 2.1 across the census of compact CY_3 is simple: the *less* fibration structure, the fewer strata the geometry occupies, the fewer gluing cocycles are available. The quintic is the minimal case (only MNOP categorical CoHA, no chart-wise pieces); $K3 \times E$ is the maximal case (four strata intersect, four cocycles assemble). The generic compact CY_3 is strictly between.

3. PARTIAL TRANSFER ON COMPACT CY_3

Although the chart-wise program of Parts II–VII terminates at the toric boundary, three pieces of the CoHA machinery transfer without modification to any compact CY_3 . Identifying what survives is the first step towards compact- CY_3 -native CoHA constructions.

Proposition 3.1 (Stratum-free machinery). **[Proposition]** *Let X be any smooth projective CY_3 (compact, not assumed toric or fibred). The following machinery transfers without modification:*

- (1) *The Davison–Meinhardt critical cohomological Hall algebra $\text{CoHA}_{\text{crit}}(X) := H_{\text{crit}}^{\bullet}(\mathcal{M}_X^{ss}, \phi_W)$, defined for any CY_3 category $D^b(\text{Coh } X)$ with symmetric obstruction theory, equipped with its Hall product via the CoHA functor $\boxtimes$ — primary source Davison–Meinhardt [arXiv:1601.02479](#).*
- (2) *The Kontsevich–Soibelman motivic wall-crossing formula in the motivic Hall algebra, with its quantum-dilogarithm cocycle and chamber-to-chamber automorphism — primary source [arXiv:0811.2435](#).*
- (3) *The two-stage factorisation $\Phi_3 = \text{Sp}_{\Sigma_2, C}^{\text{ch}} \circ \Phi_3^{FA}$, with Stage-1 canonical up to contractible choice by Kontsevich–Tamarkin formality and Costello–Gwilliam–Li locality, and Stage-2 specialising to a curve C via Dunn–Lurie $E_3 = E_1 \otimes E_2$.*

Sketch. Each item’s definition requires only the CY_3 Serre structure $\omega_X \simeq \mathcal{O}_X$ and either a symmetric obstruction theory (items 1 and 2) or an E_3 -factorisation structure arising from the three real-dimensional covering of X (item 3). These hold for every CY_3 ; no toric atlas, no BCOV vanishing, no torus action, no finite quiver is involved.

Davison integrality (item 1) produces a BPS Lie algebra $\mathfrak{g}_X^{\text{BPS}}$ graded by the numerical Grothendieck group $K^{\text{num}}(X)$; its PBW basis is indexed by BPS states regardless of whether the geometry admits chart-wise presentation. The proof in [arXiv:1601.02479](#) is ∞ -categorical and uses no toric localisation.

Motivic wall-crossing (item 2) is formulated in the motivic Hall algebra $\text{MH}(\mathcal{M}_X^{ss})$, with quantum-dilogarithm variables $\Psi(x_\gamma)$ indexed by numerical classes. The formula $\prod_{\text{incoming}} \Psi(x_\gamma)^{\Omega(\gamma)} = \prod_{\text{outgoing}} \Psi(x_\gamma)^{\Omega(\gamma)}$ holds chamber-by-chamber for any CY_3 with stability condition; no chart-wise input.

Two-stage factorisation (item 3) is constructed categorically: Stage-1 is Kontsevich–Tamarkin E_3 -formality of $\text{HH}^{\bullet}(\text{Coh } X)$ plus Costello–Gwilliam–Li holomorphic locality; Stage-2 is specialisation to a curve C — both are universal constructions requiring no geometry-specific input beyond the CY_3 condition. $\square$

Proposition 3.2 (What fails to transfer). **[Proposition]** *The following chart-wise machinery does not transfer to a generic compact CY_3 :*

- (1) *Per-chart Schiffmann–Vasserot $Y^+(\widehat{\mathfrak{gl}}_1)$ stalks located at toric vertices;*
- (2) *Galakhov–Li–Yamazaki bond factors $\varphi_{a \Rightarrow b}$ attached to toric fan edges;*
- (3) *The shuffle-algebra presentation of $\text{CoHA}(X)$ via Nekrasov–Okounkov fixed-point localisation;*
- (4) *The Van den Bergh tilting bundle producing a finite quiver with potential.*

Each item requires at least one of **(C₁)**–**(C₄)**, and Theorem 1.1 rules them out. The Stage-1 factorisation Φ_3^{FA} survives in principle, but lacks the explicit local formula that toric geometry provides — so the categorical existence of $\Phi_3^{FA}(X)$ does not yield a computation.

Remark 3.3 (The meaning of *principle-level*). Proposition 3.1 item (3) is “principle-level”: the two-stage factorisation Φ_3 exists abstractly on any compact CY₃ by Kontsevich–Tamarkin plus Costello–Gwilliam–Li, but the Stage-1 output $\Phi_3^{FA}(X)$ is not computable by chart-wise formulas. Computing it requires either (a) a specific tilting-bundle presentation (which **(O₄)** blocks), or (b) a specific local chart presentation (which **(O₁)** blocks), or (c) a specific torus equivariance (which **(O₃)** blocks). None is available for the quintic.

What is available for the quintic: the *categorical* CoHA $\text{CoHA}_{\text{crit}}(X_5)$ exists by Davison–Meinhardt; its *generating function* of BPS numbers is computed by mirror symmetry; the link between the two is the Joyce–Song integration map ([arXiv:0810.5645](#)) sending motivic classes to numerical invariants. The algebra structure of $\text{CoHA}_{\text{crit}}(X_5)$ remains non-explicit.

4. WHAT IS RECOVERED ON COMPACT CY₃

The collapse of chart-wise gluing on compact CY₃ is not the end of the story. Three global pieces of structure — the BPS Lie algebra multiplicity data, the MMNS motivic generating function, and the Borchers–Gritsenko automorphic formula — recover numerical and categorical content without producing an explicit cocycle.

Proposition 4.1 (Davison BPS Lie algebra). **[Proved elsewhere]** (Davison [arXiv:1601.02479](#) Thm. A; Davison–Meinhardt [arXiv:1601.02479](#)) *For X any smooth projective CY₃ with an auxiliary potential W (or symmetric obstruction theory), there exists a BPS Lie algebra*

$$\mathfrak{g}_X^{\text{BPS}} \subset \text{CoHA}_{\text{crit}}(X)$$

graded by $K^{\text{num}}(X)$, such that:

- (1) *The PBW theorem holds: $\text{CoHA}_{\text{crit}}(X) \simeq U(\mathfrak{g}_X^{\text{BPS}})$ as graded vector space;*
- (2) *The multiplicity of $\mathfrak{g}_X^{\text{BPS}}$ in each numerical class γ is the BPS invariant $\Omega(\gamma)$.*

For the quintic, $K^{\text{num}}(X_5) = \mathbb{Z}^{H^{1,1}} \oplus \mathbb{Z} = \mathbb{Z}^2$ (rank one from $H^{1,1}$, rank one from $H^0 \oplus H^4 \oplus H^6$ collapsing to H^0 mod the Chern-character kernel). The BPS invariants $\Omega(\gamma)$ are computed by the Candelas et al. *B*-model calculation; the BPS Lie algebra structure is encoded in the generating function but not yet tabulated vertex-by-vertex (there is no finite quiver to index the vertices).

Proposition 4.2 (MMNS motivic generating function). **[Proved elsewhere]** (Morrison–Mozgovoy–Nagao–Szendrői [arXiv:1107.5017](#)) *The motivic DT partition function of a CY₃ category can be organised into an infinite product over BPS classes:*

$$Z^{\text{DT}, \text{mot}}(X) = \prod_{\gamma} \prod_{k \geq 0} (1 - \mathbb{L}^{k-d_{\gamma}/2} y^{\gamma})^{-\Omega(\gamma) \chi(L_{\gamma})}$$

for suitable line bundles L_{γ} and motivic weights.

The MMNS formula expresses the DT generating function as a motivic Euler product, with the BPS numbers $\Omega(\gamma)$ and motivic weights $\chi(L_{\gamma})$ as numerical inputs. For the conifold the formula reduces to $Z^{\text{DT}}(\mathbf{Y}) = M(-q)^2 \prod_{k \geq 1} (1 - Q(-q)^k)^k$ (Szendrői’s two-chart form). For compact CY₃ the formula is formally available but the $\Omega(\gamma)$ values are not closed-form.

Proposition 4.3 (Borchers–Gritsenko automorphic assembly). **[Proved elsewhere]** (Borchers [arXiv:alg-geom/9609022](#), Gritsenko [arXiv:math/9906190](#)) *For a lattice-polarised Calabi–Yau threefold X whose moduli sits inside a locally-symmetric period domain $\mathcal{D}/\Gamma$, there exists a Borchers lift Ψ_X of a weak Jacobi form ϕ on the polarising lattice, with*

$$\kappa_{\text{BKM}}(\mathfrak{g}_{\Psi_X}) = c_{\Psi_X}(0)/2,$$

where $c_{\Psi_X}(0)$ is the constant Fourier coefficient of the generating Jacobi form.

On $K3 \times E$ the lattice-polarisation is the Mukai lattice $\tilde{H}(K3, \mathbb{Z}) \simeq \Pi_{4,20}$ and the Borchers lift is the Gritsenko Δ_5 form of weight 5 with constant term $c_1(0) = 10$, giving $\kappa_{\text{BKM}} = 5$. For the quintic there is no period domain on which the classical Borchers construction produces a canonical lift. Hence the automorphic assembly is stratum-(iv) specific; compact CY_3 outside stratum (iv) do not inherit it.

Theorem 4.4 (What recovers, what does not). **[Proposition]** *On a generic compact CY_3 (Picard rank one, no fibration), the available machinery produces:*

- $\text{CoHA}_{\text{crit}}(X)$ (Davison–Meinhardt, categorical);
- $\mathfrak{g}_X^{\text{BPS}} \subset \text{CoHA}_{\text{crit}}(X)$ (BPS Lie algebra, graded by K^{num});
- $Z^{\text{DT}, \text{mot}}(X)$ (MMNS motivic generating function);
- $\Omega(\gamma)$ via mirror symmetry (numerical BPS invariants);
- $\Phi_3^{\text{FA}}(X)$ categorically (Stage-1 factorisation, principle-level).

Not produced:

- Chart-wise $Y^+(\widehat{\mathfrak{gl}}_1)$ stalks;
- Explicit bond-factor formulas per edge of a toric fan;
- Shuffle-algebra presentation with torus-equivariant localisation;
- Finite quiver with potential (Q, W) ;
- Borchers–Gritsenko automorphic cocycle (absent outside stratum (iv)).

In particular, the Stage-1 factorisation exists but is not computable in closed form, and the BPS Lie algebra exists but is not combinatorially presented.

Remark 4.5 (Compact CY_3 outside $K3 \times E$). For compact CY_3 that inherit a partial stratum structure — e.g., the bicubic with $\mathbb{P}^2 \times \mathbb{P}^2$ symmetry (partial Stratum (iii)) or the Schoen threefold with two elliptic fibrations (partial Stratum (ii/iv)) — the recovery is intermediate: more than the quintic, less than $K3 \times E$. The automorphic cocycle may survive in weakened form; the chart-wise stalks do not.

Categorical CoHA via MNOP + Davison integrality is the irreducible minimum available on every compact CY_3 . It produces the BPS Lie algebra graded by K^{num} , which is the one universally accessible object. Everything beyond requires at least one of the four features, or the presence of lattice-polarisation.

5. THE $K3 \times E$ POSITIONING

$K3 \times E$ is atypical on three counts relative to the generic compact CY_3 :

- (1) It has $h^{1,0}(K3 \times E) = h^{1,0}(K3) + h^{1,0}(E) = 0 + 1 = 1$, so $\text{Aut}^0(K3 \times E) \supset E$ is a positive-dimensional algebraic group. Stratum (ii) (reduced $\mathbb{C}^\times \times \text{Aut}(X)$) is non-trivial.
- (2) It admits the $K3$ elliptic fibration $K3 \rightarrow \mathbb{P}^1$ pulled back to $K3 \times E \rightarrow \mathbb{P}^1 \times E$, giving 24 singular fibres of type $I_1 \times E$ over the 24 discriminant points. Stratum (i) (toric-like local structure) is partial.
- (3) Its periods sit in a 20-parameter locally-symmetric domain $\mathcal{D} = \text{O}(2, 19)/\text{O}(2) \times \text{O}(19)$ ($K3$ factor) times $\mathbb{H}$ (E-factor), admitting the Gritsenko Δ_5 Borchers lift. Stratum (iv) is fully available.

This triple alignment — positive-dimensional Aut^0 , elliptic fibration structure, lattice-polarised period domain — is exceptional. It is what permits the 24-curve-stalk assembly of Part VIII. The generic compact CY_3 (quintic, bicubic, Schoen) realises at most one of the three.

Remark 5.1 (Serre-equivariant quasi-NCCR, reprise). Per the CLAUDE.md key-facts, $K3 \times E$ admits *no global NCCR* due to five obstructions: (a) the dualising sheaf $\omega_{K3 \times E} = \mathcal{O}$ is not reflexive-tilting; (b) derived McKay requires finite Aut fixing a point; (c) HPD self-duality is not compatible

with product polarisation; (d) Mukai vanishing fails off the K3 factor; (e) no global CY-3 symmetric obstruction theory. The substitute is the *Serre-equivariant quasi-NCCR*: a locally-defined tilting object equivariant under the Serre twist, gluing via the factorisation pushforward.

Even with the exceptional triple alignment, $K3 \times E$ does not admit a global tilting object; the categorical CoHA is assembled through the stratum-(iv) Borchers cocycle rather than through a finite quiver. The quasi-NCCR is quasi precisely because $(\mathbf{O}_4)$ does not fully dissolve — it only bends.

6. THE 2.5-OBSTRUCTION COUNT AND THE GLUING LIMIT

The two-and-a-half-obstruction formulation of Theorem 1.1 is structural: $(\mathbf{O}_1)$ and $(\mathbf{O}_3)$ are two statements of toric rigidity (absence of a T^3 -action on either the total space or on moduli), $(\mathbf{O}_4)$ is the categorical shadow of that rigidity (no finite tilting-quiver), and $(\mathbf{O}_2)$ is a Hodge-theoretic obstruction independent of toric considerations.

Proposition 6.1 (Structural logical content). **[Proposition]** *The four obstructions reduce as follows:*

- $(\mathbf{O}_1)$ (no toric atlas) and $(\mathbf{O}_3)$ (no torus equivariance on simples) are two faces of the same statement: a smooth projective CY₃ of Picard rank ≥ 1 admits no algebraic torus of rank $\geq \dim$ acting with isolated fixed points.
- $(\mathbf{O}_4)$ (no finite quiver with potential) follows from $(\mathbf{O}_3)$: a finite quiver would produce a torus equivariance on moduli.
- $(\mathbf{O}_2)$ (BCOV anomaly) is independent: it is a statement in $H^{0,1}(X, T_X)$, which depends on the middle Hodge structure of X , unrelated to torus or quiver structure.

Hence the four named obstructions carry two-and-a-half units of independent logical content: the toric-rigidity axis $(\mathbf{O}_1) \Leftrightarrow (\mathbf{O}_3) \Rightarrow (\mathbf{O}_4)$ (counting as 1.5 because of $(\mathbf{O}_4)$'s additional categorical content) and the BCOV axis $(\mathbf{O}_2)$ (counting as 1).

The rhetorical convenience of “four” is preserved; the structural logic is 2.5. Both formulations appear in the literature and in Vol III; the synthesis point of Wave 4 Agent 8 is that any future extension of the chart-wise program should attack the axes, not the four features individually.

Remark 6.2 (What an extension would look like). A hypothetical extension of chart-wise gluing to compact CY₃ would need to address both axes:

- (1) The toric-rigidity axis: construct a *virtual torus action* on moduli via formal stacky enhancements (e.g., Stratum-(iv) Borchers-lift as an $O(2, n)$ -equivariance on a formal neighbourhood of the period domain), replacing the toric T^3 with an automorphic cocycle.
- (2) The BCOV axis: construct a *BCOV-anomaly-cancelling counter-term* in the factorisation-algebra formalism, either through Costello–Li obstruction-theoretic means or via a derived-stack enhancement that absorbs α_{BCOV} into the structure cocycle.

Each axis requires non-trivial technology, both axes simultaneously even more so. The current state is: categorical CoHA exists (Davison–Meinhardt, stratum-free); chart-wise CoHA does not (obstructed by both axes).

7. SUMMARY

We have:

- (1) Stated and derived from first principles the four obstructions $(\mathbf{O}_1)$ – $(\mathbf{O}_4)$ to chart-wise gluing on a compact CY₃ (Theorem 1.1), including the Hodge computation for $(\mathbf{O}_2)$ and the toric-rigidity argument for $(\mathbf{O}_1) \Leftrightarrow (\mathbf{O}_3)$.
- (2) Observed that the four-item enumeration collapses structurally to two-and-a-half units of logical content (Proposition ??), per Wave 4 Agent 8's rectification.

- (3) Enumerated compact CY_3 examples (quintic, bicubic, Schoen) that lack $K3 \times E$'s exceptional triple alignment and hence admit no chart-wise assembly (Section 2).
- (4) Identified the stratum-free machinery (Davison–Meinhardt critical CoHA, Kontsevich–Soibelman motivic wall-crossing, two-stage factorisation in principle) that transfers to any compact CY_3 (Proposition 3.1).
- (5) Identified the chart-wise machinery (Schiffmann–Vasserot stalks, GLY bond factors, shuffle algebras, finite quivers) that does not (Proposition 3.2).
- (6) Recorded what is recovered on a generic compact CY_3 (BPS Lie algebra, MMNS motivic generating function, Borchers–Gritsenko automorphic formula *only when stratum (iv) is present*) and what is not (chart-wise stalks, bond factors, shuffles, quivers, the automorphic cocycle outside stratum (iv)) (Theorem ??).
- (7) Positioned $K3 \times E$ as the exceptional case where all three structural features (positive-dimensional Aut^0 , elliptic fibration, lattice-polarised period domain) align simultaneously (Section ??), and noted that even there the gluing is through the Serre-equivariant *quasi*-NCCR rather than a true NCCR.

The gluing machinery of Parts II–VIII is sharp: it covers the $K3 \times E$ case and the toric- CY_3 -without-compact-4-cycles family, but not the generic compact CY_3 . The boundary is exact, dictated by the four obstructions, and any extension beyond requires engaging the two structural axes (toric rigidity, BCOV anomaly) with non-classical machinery.